FAMA-Band hub — hierarchical root. All inter-sheet nets are global labels.

Power

File: power.kicad_sch

Power: USB-C, charger, battery, 3V3 buck-boost, AFE supply + USB interlock

Analog front end

File: afe.kicad_sch

Analog front end: ADS1298, 2.048 MHz clock, input filters, electrode connectors

MCU

File: mcu.kicad_sch

MCU: ESP32-S3 module, IMU, microSD, buttons, LEDs

Generated by hardware/gen/gen_sch.py from the Python netlist — edit the netlist, not this file

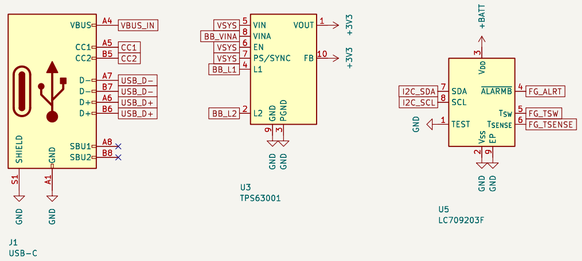
FAMA-Net thesis hardware

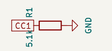
Sheet: /
File: hub.kicad_sch

Title: FAMA-Band hub — Power

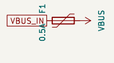
Size: A3	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Power: USB-C, charger, battery, 3V3 buck-boost, AFE supply + USB interlock

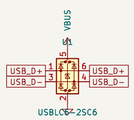




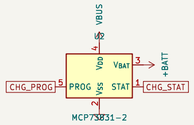
















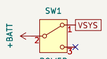














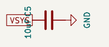


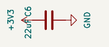


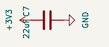


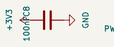






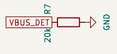


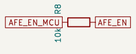


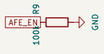


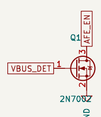




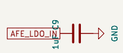




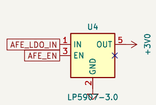




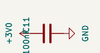


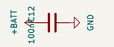


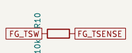
















USB-C is for charging, firmware and data offload ONLY while the band is OFF the arm.
Q1 holds the AFE LDO in shutdown whenever VBUS is present, independent of firmware,
so electrodes are never energised while a mains-referenced USB host is attached.

Generated by hardware/gen/gen_sch.py from the Python netlist - edit the netlist, not this file

FAMA-Net thesis hardware

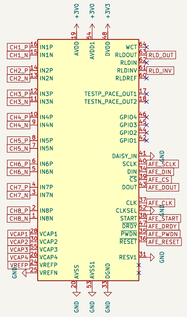
Sheet: /Power/

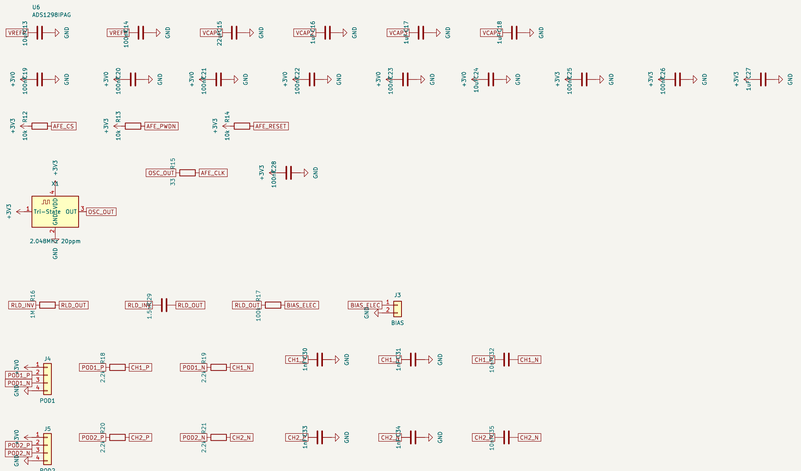
File: power.kicad_sch

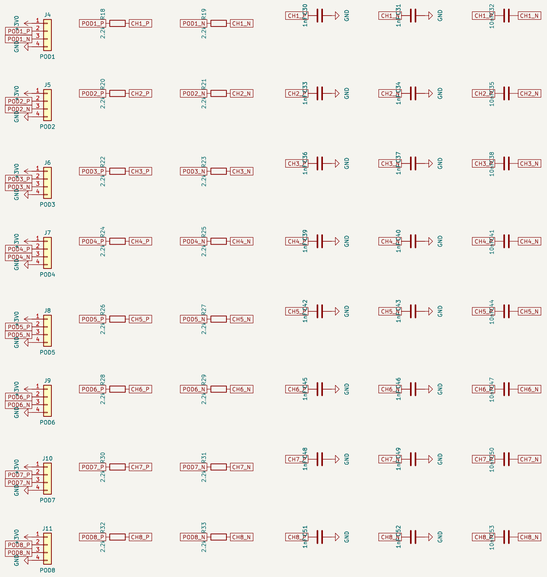
Title: FAMA-Band hub - Power

Size: A2	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Analog front end: ADS1298, 2.048 MHz clock, input filters, electrode connectors





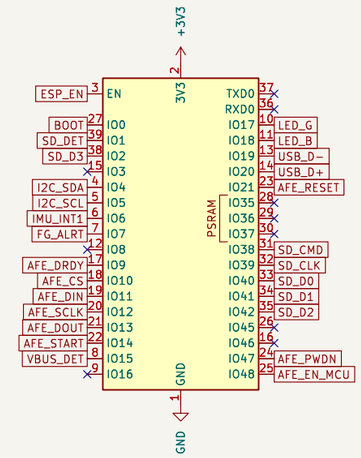


ADS1298 runs unipolar: AVDD = +3V0 (LP5907), DVDD = +3V3, VREF = 2.4 V internal.
Clock: external 2.048 MHz +/−20 ppm (CLKSEL = 0) -> f_s = 1000.00 Hz exactly (DR = 101, HR mode).
Channel rc pod buffer -> 2.2k + 1n (CM) / 10n (DM) EMI filter -> INnP/MnN; PGA gain 6,
range +/−400 mV, 1 LSB = 47.7 nV. Right-leg drive: RLDOUT -> 100k -> bias electrode.

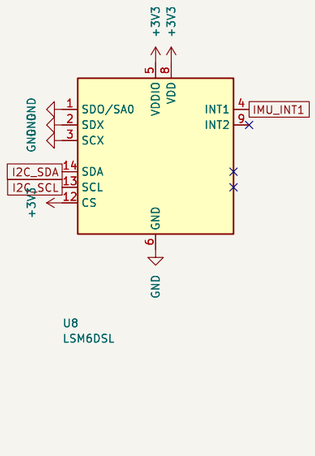
Generated by hardware/gen/gen_sch.py from the Python netlist — edit the netlist, not this file

FAMA—Net thesis hardware	
Sheet: /Analog front end/	
File: afe.kicad_sch	
Title: FAMA—Band hub — Analog front end	
Size: A1 — Date: 2026-10-25	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 3/4

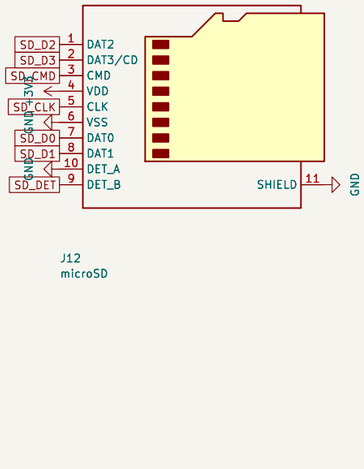
MCU: ESP32-S3 module, IMU, microSD, buttons, LEDs



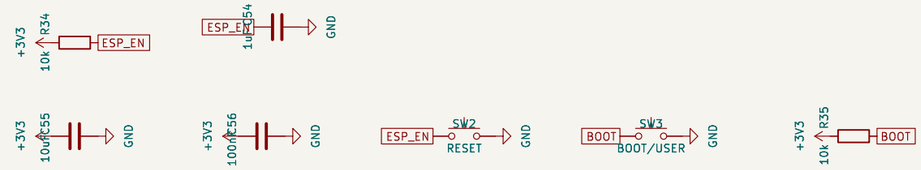
U7
ESP32-S3-WROOM-1-N16R8

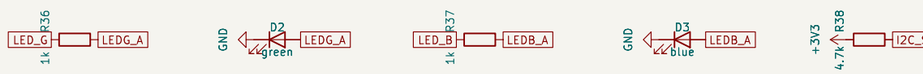


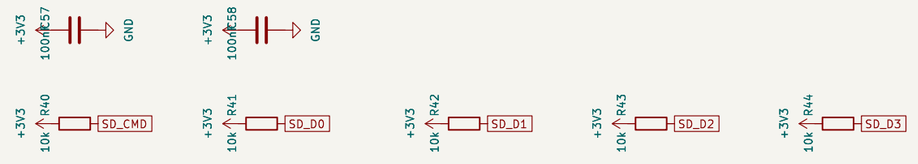
U8
LSM6DSL



J12
microSD







ESP32-S3-WROOM-1-N16R8 (octal PSRAM: GPIO35–37 reserved). Strapping pins 0/3/45/46 left at defaults.
AFE on SPI2 (GPIO9–14,21,47,48), microSD 4-bit SDMMC (GPIO1,2,38–42), I2C IMU + fuel gauge (GPIO4/5).

Generated by hardware/gen/gen_sch.py from the Python netlist — edit the netlist, not this file

FAMA-Net thesis hardware

Sheet: /MCU/
File: mcu.kicad_sch

Title: FAMA-Band hub — MCU

Size: A3	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4